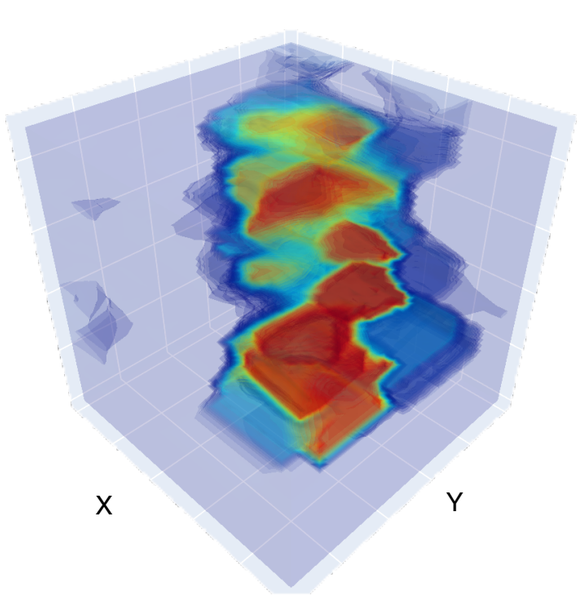
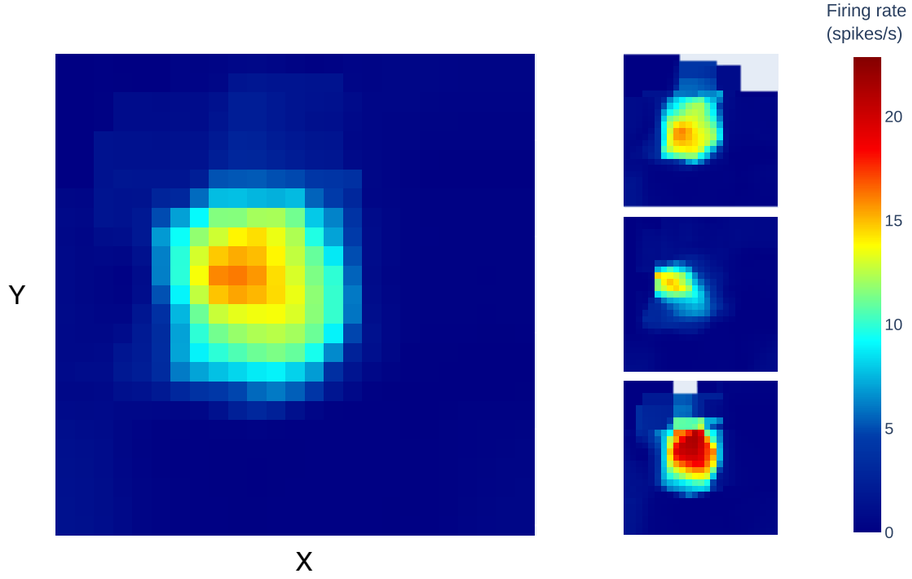
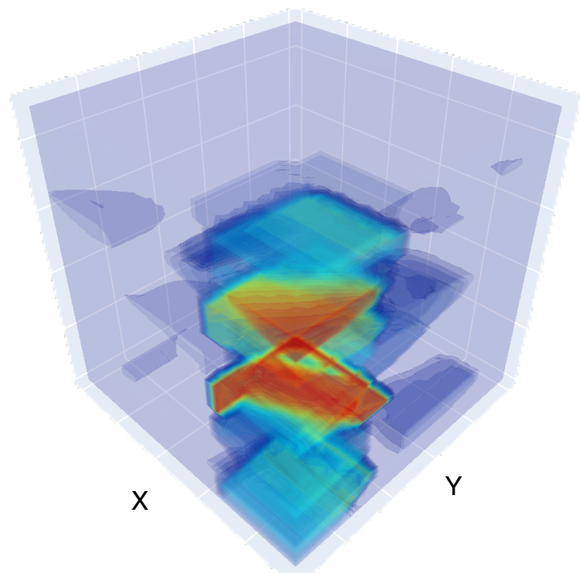
a

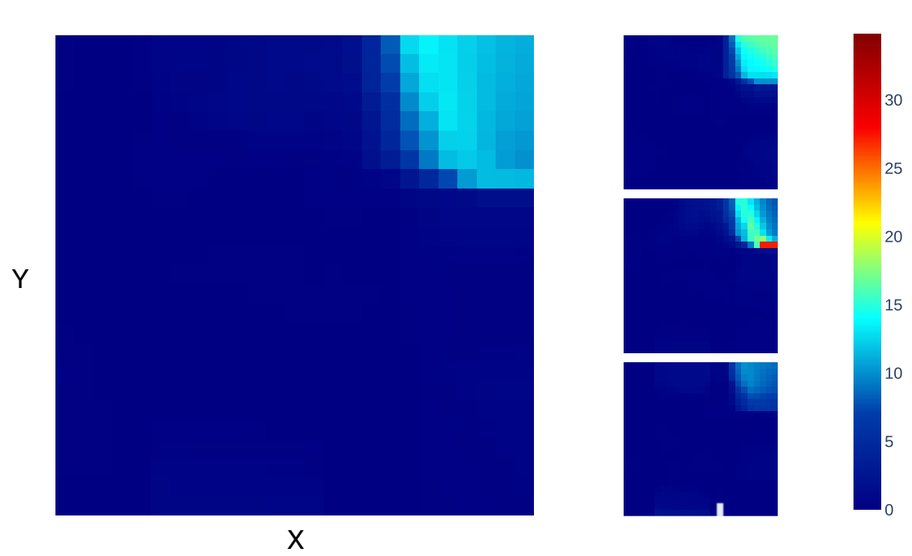

X
Y


Y
X

Firing rate
(spikes/s)
20
15
10
5
0

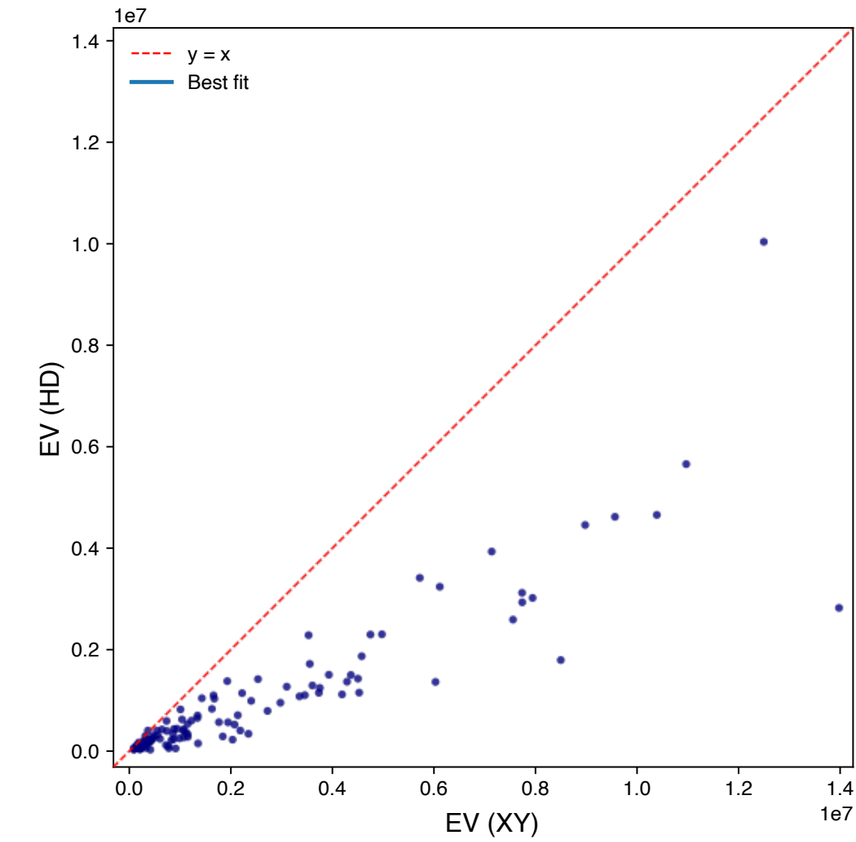
b


X
Y

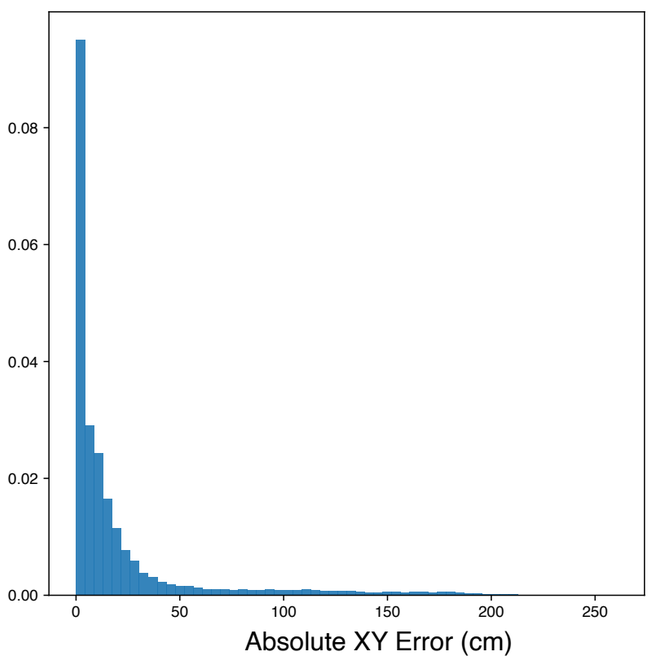

Y
X

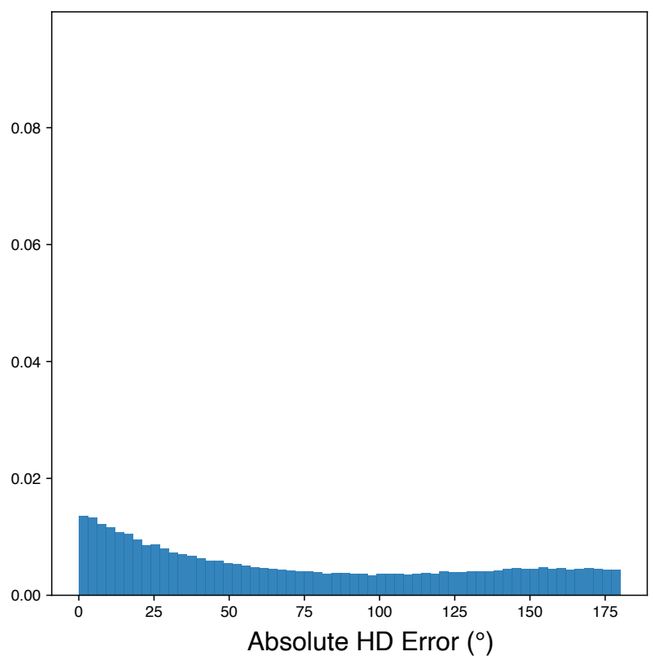
30
25
20
15
10
5
0

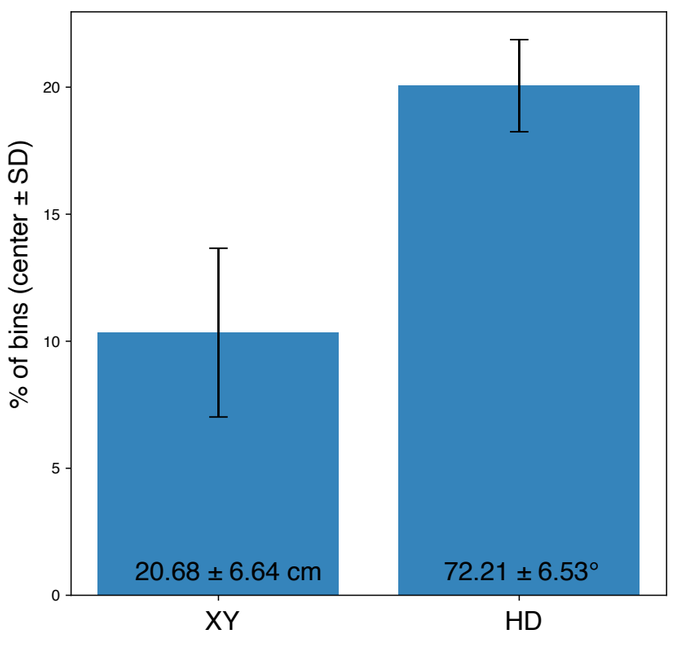
c


1e7
y = x
Best fit
EV (HD)
EV (XY)
1e7
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

d


Absolute XY Error (cm)
0.00
0.02
0.04
0.06
0.08
0
50
100
150
200
250


Absolute HD Error (°)
0.00
0.02
0.04
0.06
0.08
0
25
50
75
100
125
150
175


% of bins (center ± SD)
0
5
10
15
20
20.68 ± 6.64 cm
72.21 ± 6.53°
XY
HD